Course: CS-301 Operating Systems
Unit 4: Memory Management - Lecture 1

1. Introduction to Virtual Memory
Virtual memory is a memory management technique that provides an idealized abstraction of the storage resour

Primary Advantages:
- Large Address Spaces: Programs can be larger than physical memory (RAM).
- Protection & Isolation: Each process has its own isolated virtual address space.
- Efficient Resource Utilization: Code and data are loaded into RAM on demand (Demand Paging), reducing mer

2. Paging Mechanism
Paging is a memory management scheme by which an operating system stores and retrieves data from seconda
- Logical Address Structure: Divided into two parts: Page Number (p) and Page Offset (d).
- Physical Memory: Divided into fixed-size blocks called Frames.
- Page Size: Typically 4 KB (4096 bytes), determined by the hardware architecture.
- Page Table: A per-process data structure that maps virtual page numbers (VPN) to physical frame numbers (P

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3. Address Translation and Page Tables
When a CPU generates a virtual memory address, the Memory Management Unit (MMU) translates it to a physic
- Linear Page Table: Simple array indexed by page number. Suffers from high memory overhead in 64-bit systen
- Hierarchical (Multi-level) Paging: Breaks the page table into multiple levels (e.g., outer page table, inner page ta
- Inverted Page Table: Has one entry for each real page (frame) of memory, rather than each virtual page.

4. Translation Lookaside Buffer (TLB)
Because page tables reside in main memory, every memory reference requires at least two physical memory ac
To alleviate this bottleneck, hardware provides a special, fast-lookup hardware associative cache called the Tran
- TLB Hit: The page number is found in the TLB. Translation occurs in a single clock cycle.
- TLB Miss: The page number is not in the TLB. The MMU or OS walks the page table in RAM, loads the entry in
- Effective Memory Access Time (EMAT) formula:
  EMAT = (Hit Ratio * TLB Access Time) + ((1 - Hit Ratio) * (TLB Access Time + 2 * Memory Access Time))

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5. Page Replacement Algorithms
When a page fault occurs and all physical memory frames are allocated, the operating system must choose a vic

Major Algorithms:
A. FIFO (First-In, First-Out):
- The simplest page-replacement algorithm. When a page must be replaced, the oldest page is chosen.
- Belady's Anomaly: For some page-replacement algorithms, the page-fault rate may increase as the number of

B. Optimal Page Replacement (OPT / MIN):
- Replaces the page that will not be used for the longest period of time in the future.
- Has the lowest possible page-fault rate of all algorithms.
- Theoretical benchmark: Impossible to implement in practice because it requires future knowledge of reference s

C. LRU (Least Recently Used):
- Replaces the page that has not been referenced for the longest period of time in the past.
- Approximates the optimal algorithm by using past behavior to predict future accesses.
- Implementation methods: Counter (time-of-use register) or Stack (doubly linked list moved to top on each refere
- Does NOT suffer from Belady's Anomaly (it belongs to the class of stack algorithms).

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6. Thrashing and System Degradation
Thrashing occurs when a computer system spends more time servicing page faults and swapping pages in and o

Causes of Thrashing:
- High degree of multiprogramming: If the OS introduces too many active processes, the number of frames alloca
- Cascading page faults: When a process faults, it queues for the paging disk. The CPU utilization drops becaus

7. Solutions to Thrashing
A. Working-Set Strategy:
- Based on the principle of Locality of Reference (spatial and temporal locality).
- Working Set Window (Delta): A fixed parameter representing a number of recent page references.
- If total memory demand of all working sets exceeds total physical memory available, the OS suspends one or m

B. Page Fault Frequency (PFF) Control:
- Measures page fault rate directly.
- If page fault rate exceeds an upper threshold, allocate more frames to the process.
- If page fault rate drops below a lower threshold, reclaim unused frames.